# Modeling Current and Future Carbon Sequestration Potential of Trees in a Defined Area

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**Abstract**

This project aims to calculate the carbon sequestration potential of areas of high green cover in The Hyderabad Public School, Begumpet, using modeling techniques. In computing the carbon absorption and storage capacity, vegetation cover in a region of similar geography has been used to compute tree density, which, combined with approximations about individual tree biomasses, enables the creation of an estimate of total sequestration capacity. This, combined with analysis of local urban sprawl to confirm the role of the school to protect trees and prove additivity, further legitimizes carbon credit generation potential. Finally, predicting future carbon credit prices using differential equations provides an approximation for revenue that can be generated from selling credits. Generalizing this process to other sizable private areas shows the important role such private land holdings can play in growing global carbon markets.

## 1 Introduction

Carbon Markets are carbon pricing mechanisms that enable government and non-government actors to sell, generate and trade carbon credits.[1] Carbon credit are financial entities that represent the right to generate or emit 1 tone of a greenhouse gas, typically carbon dioxide.[2] These credits are generated when steps are taken to sequestrate the said amount of carbon dioxide from the atmosphere, either through ventilators that absorb the gas or through planting and maintain trees. Article 6 of the Paris Agreement of 2015 lays out the provision for a crediting mechanism, which by allowing entities to transfer credits enables financial methods for such entities to reach net-zero targets. [3] Additionally, cash flow generated though this transaction can be used to further pursue efforts against climate change and global warming. Currently, there are 75 countries in total that operate carbon pricing instrument, namely carbon taxes and emissions trading systems (ETS), with India being a country that is strongly considering the imposition of such a policy.[4] Currently, the government expects to start a local ETS by mid-2026, thus, providing an opportunity

for organization in the meantime to survey land with sizable vegetation to estimate sequestration potential and generate carbon credits.

The most accurate and common method to determine carbon sequestration is to perform a survey of trees in the defined land area. This method aims to calculate the mass of a tree (m) using various physical characteristics such as girth circumference (C) and height (h). Wood density ($\rho$) is an important variable determinant in this calculation, but using a sample from tropical evergreen and deciduous trees in Tamil Nadu, a geographically and ecologically similar region, indicates that mean wood density is $740120 kg/m^3$.[5] Using this estimate, a generalized formula for dry weight, which is 50% of total weight, can be computed as follows:

$$m = \rho \cdot V$$
$$m = \rho \cdot \pi \cdot r^2 \cdot h$$
$$m = \rho \cdot \frac{C^2}{4\pi} \cdot h$$
$$m_{dry} = 0.5 \cdot \rho \cdot \frac{C^2}{4\pi} \cdot h$$
$$m_{dry} = 29.05 C^2 h$$

A combination of simple measurements and trigonometry can be used to determine the carbon storage potential of trees using this method; however, such a method is cumbersome and can take a very long time to implement. The method suggested in this essay provides an easily obtainable, albeit slightly less accurate, estimate of this number using modeling techniques.

# 2 Modeling

There are three modeling problems in this project: 1) What is a reliable estimate for $CO_2$ sequestered by trees on the campus? 2) How can it be quantitatively proved that the campus presents a significant additional contribution in carbon capture? And 3) What are the projected future revenue flows from this source? This section aims to answer these questions.

## 2.1 Carbon Sequestration Potential

Tree density, defined as the number of trees per unit area, is an important determinant in calculating the number of trees, hence the carbon sequestration potential of a given area. Tree density ($\tau$) roughly is based on geographical factors (like latitude ($L$)), climactic factors (like precipitation ($P$) and temperature ($T$)), and environmental factors. Assuming there is no artificial deforestation and assuming that in the long term natural disasters are compensated for by

extensive periods of vegetative propagation, tree density can roughly be given by the below proportionality:

$$\tau \propto \frac{T_{avg} \cdot P_{avg}}{L}$$

For two small areas in the same region, these factors are likely to be nearly identical; hence, tree density can be assumed to be the same. Further, analysis of the green spectral density at both HPS and KBR National Park confirmed similar tree densities. Hence, the tree density of the KBR Park, of 150 trees per acre, serves as a good approximation of the tree density in selected regions of the Hyderabad public school.[6]

As the school has parts that have high vegetation while other areas are sparse, therefore it is essential to determine which areas are green and which areas are not. This can be done by passing it through a simple filter, which converts the image into RGB (red, green, blue) values. The area in the image was classified as green based on the following logic:

$$GREEN = \begin{cases} TRUE : B + G > 100 \\ FALSE : B + G < 100 \end{cases}$$

This tolerance was necessary as satellite imagery often has darker shades of green, which have more blue pixels than green pixels. To now compute the area, the haversine formula needs to be used due to the spherical shape of the earth. Distance between two points on earth due to its circular cross-section can be given as $R_{EARTH}\theta$. While radius of the Earth is a standard value, $\theta$ needs to be calculated for the given latitude ($x$) and longitude ($y$) using the haversine formula.[7] Area by dividing it in to rectangles, of known width and breadth computed using the above formula.

$$D = 2R_{EARTH} \arcsin \sqrt{\sin^2(\frac{\Delta x}{2}) + \cos y_1 \cos y_2 \sin^2(\frac{\Delta y}{2})}$$

Figure 1 represents all the areas of the campus that do qualify as areas of high vegetation (marked in white) based on our definition and have capacity for carbon sequestration. This comes to a total of 50.55 acres. Multiplying the tree density by this number, it can be estimated that the total number of trees in these regions is 7583.

The next step in estimating carbon sequestration capacity is calculating the total biomass and dry weight of the trees. This, as mentioned earlier, is dependent on the circumference and the height of the trees. Using a sample size of trees to determine the confidence interval of the population height and circumference. Tables 1 and 2 indicate the sample data consisting of 10 trees:
As the size of the population mean exceeds 30, according to the Central Limit Theorem (CLT), the values of the tree height and circumference, a t-test can be




Figure 1: Areas with Carbon Sequestration Potential

Table 1: Height

| Category | Value |
|---|---|
| Mean ($m$) | 18.63 |
| Standard Deviation ($m$) | 2.43 |

Table 2: Girth Circumference

| Category | Value |
|---|---|
| Mean ($m$) | 1.40 |
| Standard Deviation ($m$) | 0.12 |

used to determine confidence interval. Taking a 95% confidence interval, using the formula given below to determine a range of these two variables:

$$CI = \bar{x} \pm t_{0.975,9}\sigma$$

The range for height is [13.22, 24.04], and for circumference is [1.13, 1.67]. Random values in this given range can be substituted into the dry weight formula to calculate the total amount of carbon dioxide sequestered. The dry mass of the tree and the amount of carbon dioxide sequestered from the atmosphere are related by the below formula, assuming 1 kg of carbon produces 3.67 kg of $CO_2$:

$$Amt\ of\ CO_2 = 0.5 \cdot 3.67 \cdot m_{dry}$$

Using this formula, the total amount of $CO_2$ sequestered in the entire lifetime of all trees is 14941236 kg. The average age of most species of trees is around 50 years. When this value is divided by 50, the annual carbon sequestration capacity is obtained to be 298,824 kg. As this is purely an educated estimate, sensitivity analysis in the figure below indicates a possible range of annual sequestration capacity. Considering that the average lifetime of a tree is 50 years, the annual number of carbon credits generated by trees in the school is approximately 299 credits. This is equal to the amount of carbon dioxide produced by driving 2.13 million kilometers and presents a significant number of carbon credits to monetize.

| Average Lifetime of a Tree (years) | Annual Sequestration Capacity (Credits) |
|---|---|
| 30 | 498 |
| 40 | 374 |
| 50 | 249 |
| 60 | 213 |

Table 3: Estimated Carbon Sequestration Potential

## 2.2 Proving Additivity

Additivity in the context of carbon sequestration is an important parameter to ensure that the project is presenting an additional reduction in carbon emissions. This is necessary to ensure that the carbon offset is 'real' and 'credible' and wouldn't have otherwise been performed naturally.[8] In order to prove it for the proposed carbon sequestration project, it is necessary to model the urban sprawl in the same area if not for the presence of the campus.

Population density varies inversely with the distance from the city center, as pointed out by Clarke et al. in giving the following formula with positive constants $A$ and $b$ to represent population density at the center and decay per unit distance, respectively:

$$P(r) = Ae^{-br}$$

Using this formula, with Hyderabad's population density of 18,161 people per square kilometer and given that HPS is 3.48 km from the city center, the said area is bound to have a high population density. Data from the 2020 census also backs the claim, with the Begumpet region having a population density of 11,500 people per square kilometer.[9] Therefore, if not for the school, the area of 50.55 acres (0.205 sq. km) under analysis would be home to 2358 people.

Population influx over a period will lead to urban sprawl and deforestation. Didiharanyo and Kasse et al. used differential equations to model the relation between these variables, with them using industrialization as a substitute for measuring urban sprawl. They formulated the following linear first-order differential equations: [10]

$$\frac{dx}{dt} = rx(1 - \frac{x}{k}) - qx - \alpha xy - \beta xz$$

$$\frac{dy}{dt} = sy(1 - \frac{y}{L}) - py + \sigma\alpha xy + \zeta yz$$

$$\frac{dz}{dt} = \delta\beta xz + b\zeta yz - \gamma z$$

Where:

- $x(t)$: Density of Forest Resources
- $y(t)$: Human population density
- $z(t)$: Level of industrialization
- $r$, $s$: Natural growth rates of forest resources and population, respectively
- $K$, $L$: Carrying capacities for forest resources and population
- $q$: Natural degradation rate of forest resources
- $\alpha$, $\beta$: Rates of deforestation due to population density and industrialization
- $p$: Natural mortality rate of the population
- $\sigma$: Rate of population growth supported by forest resources
- $\zeta$: : Population growth rate due to migration
- $b$: Proportion of the population employed in industry
- $\delta$: Industrial growth rate influenced by population density
- $\gamma$: Governmental control coefficient over industrialization

Further, they also found equilibrium points on these differential equations, implying the existence of solutions and a direct relation between these variables. These equations show a negative relation between population growth and the density of forest resources and a direct relation between population growth and industrial growth, indicating that if the area under analysis housed over 2000 people, urban sprawl is bound to lead to deforestation.

In continuing this line of reasoning, the comparison between the sprawl within the HPS campus and that in its immediate surroundings, in Fig. 2, shows that the school is the cause for the reduced urban density. The figures below show a visual comparison of the difference in building density, which allows for the growth of the natural environment, making HPS a carbon sink in the center of the city.

## 2.3 Projecting Revenue

In July 2024, the Government of India laid out a detailed proposal for the establishment of a domestic carbon market, proposed in the Energy Conservation Act, 2022.[11] These markets, like those in Europe, will enable polluters to buy carbon credits to offset their emissions, which could become a lucrative source of additional revenue. In the European Union's ETS, a single carbon is priced at €69.14 (as of May 2025); thus, the 299 annual credits generated will generate a revenue of €20,672 or Rs. 19,74,393. (at an exchange rate of 1€ = 95.5)



Figure 2: Comparison of Urban Sprawl

The price of carbon credits is bound to be different for different markets depending on how developed the market is and government regulations surrounding carbon emissions. Europe has the highest price of 1 carbon credit due to government regulation limiting regulations and forcing companies to adopt net-zero plans.

To predict the future carbon credit price, a modified Geometric Brownian Motion can be used. The geometric Brownian motion is the solution to a stochastic differential equation used to predict the value of asset prices in the future as a function of time (t). Additionally, it also considers the current price ($S_0$), the expected rate of return ($\mu$), the standard deviation of the stock ($\sigma$) and a random error component ($\epsilon$) called the Wiener process. The geometric Brownian motion is given by the following formula:

$$S(t) = S_0 e^{(\mu - \frac{\delta^2}{2})t + \delta\epsilon}$$

Modifying this equation to also take into account government regulation ($R$) and the adoption of ETS as a common mechanism for carbon offset ($A$) gives the equation:

$$\frac{dS}{dt} = S_0 e^{(\mu - \frac{\delta^2}{2})t + \delta\epsilon} + \frac{dR}{dt} \cdot \frac{dA}{dt}$$

Qualitatively analyzing this equation indicates that carbon credit prices are going to rise as time progresses. This is going to be because of greater government regulation on carbon emitters and as ETS becomes a more popular and accessible tool. Therefore, there is a good chance that revenue generated from selling carbon will exceed even €30,000 in the future.

# 3 Conclusion

This project demonstrates the process by which mathematical modeling can be used to estimate the number of carbon credits The Hyderabad Public School can generate and further shows that it can be a profitable revenue stream. That being said, it is important to underscore that this study only presents a quantitative estimation for the number of credits generated, and for an organization to

validate the credits, a survey will have to be conducted. In the project methodology, limitations like approximation or ignorance of factors like the possibility of change in government possibility and constancy of wood density will impede accuracy, but these risks are believed to be insignificant. Carbon markets are likely to play an important role in the global fight against climate change, and the role of small players like schools or farmhouses, detailed through this project, will play a major role in their development, bringing us closer to our aims of a sustainable and risk-free planet.

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